# Edge-Based Sparse Reservoir Computing and State Observers for Real-Time Battery Diagnostics in Cyber-Physical Systems

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***Abstract*—Reliable estimation of State of Charge (SOC) and State of Health (SOH) in Lithium-Ion batteries is critical for electric vehicles (EVs) and smart grids. Traditional estimators, such as the Extended Kalman Filter (EKF), rely on high-fidelity physical models but degrade under unmodeled dynamics and cell aging. Conversely, deep recurrent neural networks present high computational costs that prevent edge deployment. This paper presents a co-designed cyber-physical system combining a 2-RC Equivalent Circuit Model (ECM) simulator, EKF state observers and Echo State Networks (ESNs) for state tracking. Additionally, we implement an optimized, edge-capable ESN classifier on an ARM Cortex-M microcontroller for thermal safety diagnostics. By introducing Compressed Sparse Row (CSR) sparse matrix-vector multiplication (SpMV) and fixed-point Q12/Q15 integer arithmetic with lookup table (LUT) linear interpolation, we achieve a **6.7× execution speedup** and save **~10 KB of Flash memory**, while maintaining classification accuracy at **98.40%** under dynamic drive cycles. An Artix-7 FPGA Verilog RTL pipeline achieves **200/200 bit-exact golden parity** with zero numerical deviation.

***Keywords*—Battery state estimation; Extended Kalman Filter; Echo State Network; Reservoir Computing; Compressed Sparse Row; Fixed-Point Arithmetic; FPGA RTL; Cyber-Physical Systems.**

## I. EXPERIMENTAL RESULTS & HARDWARE BENCHMARKS

The proposed cyber-physical framework was evaluated across synthetic 2-RC thermal stress cycles, real NASA B0005 aging datasets, and target edge microcontrollers.

| Benchmark Metric | Baseline / Float32 | Optimized / Hardware Target | Performance Improvement / Verification |
|---|---|---|---|
| C99 ARM Cortex-M Speedup | Dense Float32 (1.00x) | CSR SpMV Q15 Fixed-Point | 6.7x Execution Acceleration |
| Firmware Flash Memory Footprint | 13.4 KB (Dense Weights) | 3.2 KB (CSR Encoded) | 10.2 KB Flash Saved (~76% reduction) |
| Thermal Safety Diagnostic Accuracy | 99.80% (Float32) | 98.40% (Q15 Fixed-Point) | 1.4% drop with 0 FPU dependency |
| SOC Observer RMSE (Drive Cycles) | EKF Baseline: 1.42% | Decoupled ESN Observer: 0.80% | 43% Error Reduction |
| Xilinx Artix-7 FPGA RTL Verification | Python Golden Model | 5-Stage Pipelined Verilog RTL | 200/200 Bit-Exact States (0 Deviation) |
| Automated Pytest Regression Suite | N/A | 62 End-to-End Tests | 100% Pass Rate across all modules |

## II. SYSTEM METHODOLOGY & PIPELINE OVERVIEW

**1. Physics Simulation:** 2-RC ECM with dynamic Arrhenius thermal coupling and temperature-dependent internal impedance.
**2. Multi-Timescale Observer:** Fast SOC state estimation via EKF decoupled from slow capacity fade tracking via RLS.
**3. Sparse Leaky ESN:** Continuous-time leaky integrator reservoir with spectral radius tuning ($\rho$=0.95, $\alpha$=0.3).
**4. Microcontroller Deployment:** Compressed Sparse Row (CSR) storage format with Q15 integer arithmetic and 256-word tanh LUT.
**5. FPGA Accelerator:** Fully pipelined RTL on Xilinx Artix-7 processing sequential time-series inputs at zero float overhead.

*Note: The authoritative manuscript containing all mathematical proofs, ECM schematics, circuit figures, and code listings is maintained in markdown format at **reference/paper.md**. This document serves as the formal IEEE publication deliverable.*